

ALIAH UNIVERSITY

Name of the Examination: End Term

Session:2019-20

Subject: Engineering Chemistry

Course code: CHMUGBS01

Duration: 2 Hrs

Full Marks: 40

Semester: II

GROUP- A (Answer ANY TWO)

01. Explain the following terms: (a) Intensive properties (b) Isothermal process (c) Enthalpy (d) Reversible process (e) State function 1x5
02. Deduce the mathematical expression for the First law of thermodynamics. Explain each terms involved in the equation. 2+3
03. Derive the relations among P, V and T for the adiabatic process. Derive how C_p and C_v is related each other for ideal gas. 2.5+2.5

GROUP- B (Answer ANY TWO)

01. Explain the followings: Eutrophication and Heavy water 2.5x2
02. Write a short note on water pollution. Explain the procedure for the elimination of water pollution. 2+3
03. What causes temporary hardness of water? Explain the procedure to eliminate this. How to measure hardness of water? 1+2+2
04. What is the source of stellar energy? Present any one cycle of it. 2+3
05. Define binding energy and explain its importance with respect to atomic nucleus 5
06. Explain reverse osmosis mention and write one use of it. Write any nuclear chain reaction 2+3

GROUP- C (Answer ANY FOUR)

01. What is polymer? Write a short note on the properties of Teflon. 2+3
02. Write short notes on Conducting Polyaniline and Gutta Percha 2.5+2.5
03. Why are Silicones called Inorganic Polymers? Discuss the synthesis procedure of linear chain silicones. 2.5+2.5
04. Write short notes on the properties and preparations of Bakelite. 5
05. What is meant by vulcanization of rubber? Write its uses. 5

THE END